# Inpatient claim review assistant

### The system in one sentence

Help staff review hospital claims with checked rules, clear evidence and an owned next step.

### Key design calls

Publish drafts for staff. Rejected: Automatic settlement.. Why: Payment authority is unknown.

Code applies dated rules. Rejected: Model-made limits.. Why: Coverage must be repeatable.

Check each source quote. Rejected: Trust fluent explanations.. Why: Staff need proof.

Read back uncertain saves. Rejected: Blind retry.. Why: Avoid duplicate records.

### Who acts and who decides

| Action | Autonomy |
|---|---|
| Read approved claim files | acts alone |
| Explain evidence | suggests |
| Set eligible charges | acts alone |
| Resolve exceptions and rejections | human decides |
| Settle claims | human decides |

### Top risks by RPN

FM2 | Wrong coverage or room adjustment | RPN 225

FM8 | Human queue silently stalls | RPN 200

FM3 | Forged citation or malicious document | RPN 180

FM5 | Uncertain save lost or repeated | RPN 160

FM4 | Duplicate/concurrent drafts confuse adjudication | RPN 128

### Launch blockers and owners

H1: First release publishes staff-facing recommendations... Owner: Claims director.

H2: Insurer supplies signed, effective-dated policy... Owner: Insurance rules owner.

H3: Privacy owner selects deployment region... Owner: Privacy officer.

H4: Claims team covers a 24-hour... Owner: Claims operations lead.

### Illustrative cost per case

USD 0.02112 per case in the model budget. This is an estimate, not a current provider quote. Staff and tools cost extra.

Review status: independent-agent; round 2; pass.

Proposed design. Full rules and blockers: design.json and assumptions.md.

## Task tree

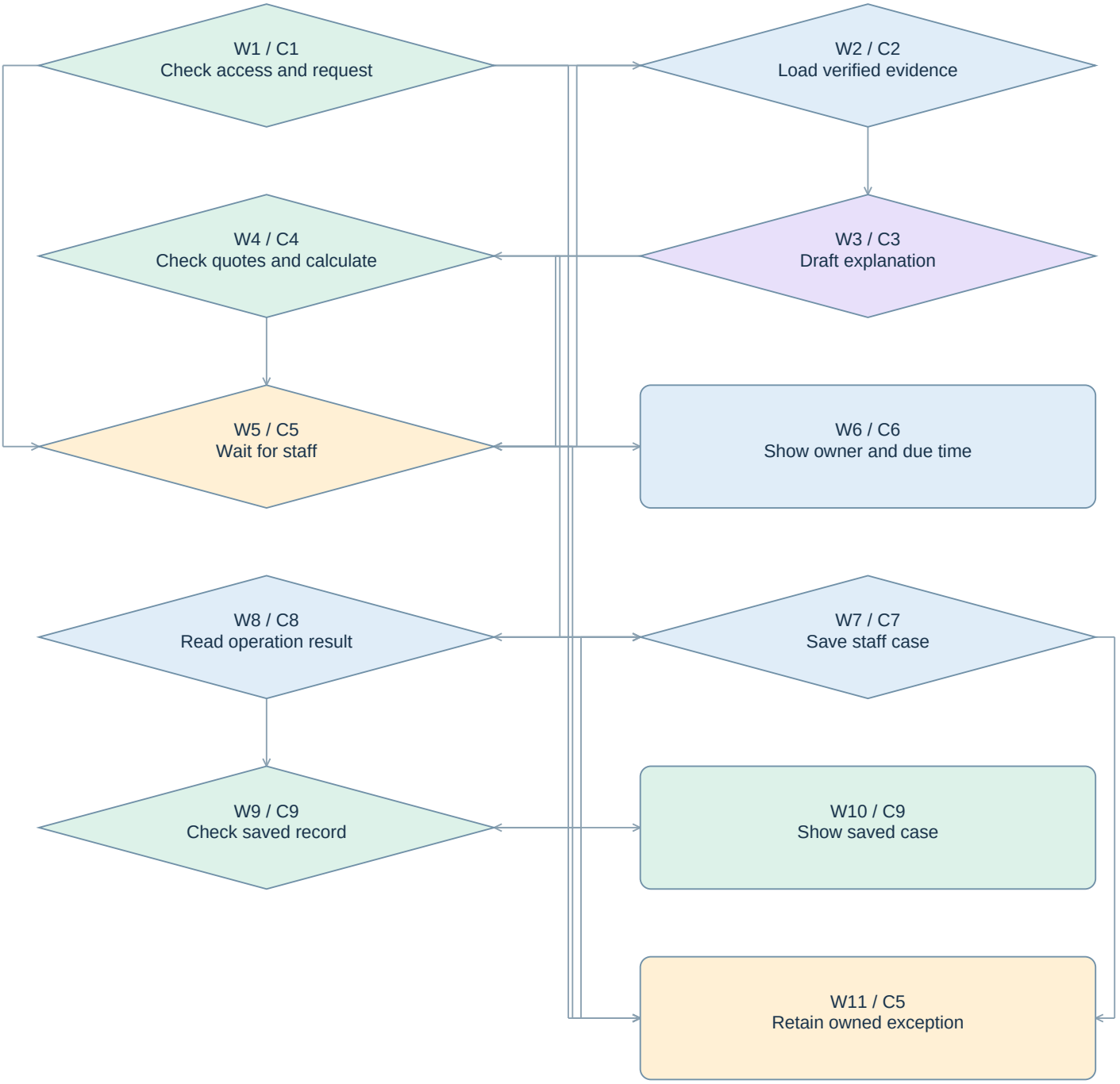
Labels show each leaf mode and risk. Full task inputs, examples and limits are in design.json and PRD.md.

## Plain words and scope

Start with draft reviews. Staff decide exceptions and rejections. All claim payments stay with the insurer.

A fingerprint is a check that the approved details have not changed. A record ID is proof from a tool that the change was saved.

## Workflow diagram




Diamonds mark choices. Arrows show all routes, including retry and recovery. Exact branch rules are in architecture.md.

| Colour | Role |
|---|---|
| #E8E0FA | Agents (LLM) |
| #E0EDF8 | Code: tools |
| #DDF2E9 | Code: validation |
| #FFF0D5 | Code: human triggers |

## Top risks and release checks

## Risk heat table - severity x occurrence

Each cell names the risks at that severity and occurrence. The full scores, causes, controls and owners are in fmea-register.csv.

| S / O | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| 10 | - | - | FM1 | - | - | - | - | - | - | - |
| 9 | - | - | FM7 | - | FM2, FM3 | - | - | - | - | - |
| 8 | - | - | FM6 | FM4, FM5 | FM8 | - | - | - | - | - |
| 7 | - | - | - | - | - | - | - | - | - | - |
| 6 | - | - | - | - | - | - | - | - | - | - |
| 5 | - | - | - | - | - | - | - | - | - | - |
| 4 | - | - | - | - | - | - | - | - | - | - |
| 3 | - | - | - | - | - | - | - | - | - | - |
| 2 | - | - | - | - | - | - | - | - | - | - |
| 1 | - | - | - | - | - | - | - | - | - | - |

| Risk | Control summary / full register linked |
|---|---|
| FM1 Wrong subject medical disclosure | Deny before medical fetch; row-level tenant/member checks and source membership |
| FM2 Wrong coverage or room adjustment | Signed effective-date precedence, explicit all required rules, integer arithmetic; unknown rules withhold amount Require C2 authoritative accumulator snapshot with same-request version... |
| FM3 Forged citation or malicious document | Source-visible OCR admission; exact anchored quote validation; contradictions route review |
| FM4 Duplicate/concurrent drafts confuse adjudication | Submission identity uniqueness, CAS, semantic candidates require linking |
| FM5 Uncertain save lost or repeated | Independent readback service, frozen intent, bounded reads; no blind retries |
| FM6 False completion without proof | Readback verifies case/operation/version and stored bytes before saved display |
| FM7 Revoked authority still writes | Fresh writer-side RBAC and snapshot version checks; no payment credentials |
| FM8 Human queue silently stalls | Provisional intake; one-hour pull status; four-hour escalation; 24-hour owned overdue state |

Risk scores are proposed estimates. They are not measured odds or risk acceptance.

## Components and contracts

This inventory is a summary. Exact typed inputs, outputs and permission rules are in architecture.md and the build milestone files.

| Component / group | Responsibility | Task IDs |
|---|---|---|
| C1 / Code: validation | Authorize intake | T1 |
| C2 / Code: tools | Verify source evidence | T2 |
| C3 / Agents (LLM) | Explain evidence | T3 |
| C4 / Code: validation | Validate policy and calculate | T3, T4 |
| C5 / Code: human triggers | Own human review | T5, T7 |
| C6 / Code: tools | Show waiting status | T5 |
| C7 / Code: tools | Save staff-facing case | T6 |
| C8 / Code: tools | Reconcile save | T6, T7 |
| C9 / Code: validation | Guard record proof | T7 |

## Shared control boundary

Code checks identity, money, dates and limits. A model may suggest a choice. It cannot authorise or perform a business write.

## Architecture and model budget

### F1 - Durable code state machine

Choice: Database-backed durable workflow and one stateless explanation model

Rejected: Unbounded autonomous adjudication agent; in-memory chat chain

Trade-off: Select on durable waits, audit, replay and retry control. Database workflow is enough for eleven nodes; a workflow engine is an unverified...

### F2 - Separate records and retrieval

Choice: Encrypted document vault, authoritative rules registry, relational case/operation ledger; no persistent vector index in release one

Rejected: One mixed model memory store

Trade-off: Documents establish evidence; insurer registry establishes policy. Tenant-keyed reads, row isolation, encryption, secrets vault, PHI-free telemetry; deletion/hold and region blocked on H3.

Access: Read scoped documents/rules; case.write only; independent operation.read; no payment/denial/send permission

### F3 - Narrow case writer

Choice: Read medical sources for bound tenant/member; write only staff-case operation

Rejected: Broad claims/payment API credential

Trade-off: One case revision blast radius; fresh RBAC/CAS, full-approved details, isolated reconciliation reader. Service owns unresolved operations after session expiry.

Access: Read scoped documents/rules; case.write only; independent operation.read; no payment/denial/send permission

### F4 - Provider-neutral private deployment

Choice: Choose existing approved insurer platform after H3 verification spike

Rejected: New multi-cloud deployment

Trade-off: Compare candidates on region, access boundaries, durable timers, deletion evidence and operating burden with identical tests; no current vendor claims.

These are reader summaries. Complete choices, alternatives, gates and owners are in PRD.md and design.json.

## Model cost

Illustrative model cost is USD 0.02112 per case. Unknown prices are not free service. Confirm rates, traffic, staff and tools before launch.

Full call counts, token budgets, rejected choices and costs: design.json, architecture.md and assumptions.md.

## Evaluation coverage matrix

| Risk / task | EV1 | EV2 | EV3 | EV4 | EV5 | EV6 | EV7 | EV8 | EV9 | EV10 | EV11 | EV12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FM1 | - | X | - | - | - | - | - | - | - | - | - | - |
| FM2 | X | - | X | X | X | - | - | - | - | - | - | - |
| FM3 | - | - | - | - | - | X | X | X | X | - | - | - |
| FM4 | - | - | - | - | - | - | - | - | - | X | X | - |
| FM5 | - | - | - | - | - | - | - | - | - | - | - | X |
| FM6 | - | - | - | - | - | - | - | - | - | - | - | X |
| FM7 | - | - | - | - | - | - | - | - | - | - | - | - |
| FM8 | - | - | - | - | - | - | - | - | - | - | - | - |
| T1 | X | X | - | - | - | - | - | - | - | - | - | - |
| T2 | X | X | - | - | - | X | X | X | X | - | - | - |
| T3 | X | - | - | - | - | X | X | X | X | - | - | - |
| T4 | X | - | X | X | X | X | X | X | X | - | - | - |
| T5 | - | - | - | - | - | - | - | - | - | - | - | - |
| T6 | X | - | - | X | - | - | - | - | X | X | X | X |
| T7 | X | - | - | X | - | - | - | - | - | - | - | X |

## Evaluation coverage matrix - continued

| Risk / task | EV13 | EV14 | EV15 | EV16 | EV17 | EV18 | EV22 | EV23 | EV24 | EV25 | EV26 | EV27 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FM1 | - | - | - | - | - | - | - | - | - | - | - | - |
| FM2 | - | - | - | - | - | - | X | X | X | - | - | - |
| FM3 | - | - | - | - | X | - | - | - | - | - | X | - |
| FM4 | - | - | - | - | - | X | - | - | - | - | - | - |
| FM5 | X | - | - | - | - | - | - | - | - | - | - | - |
| FM6 | - | X | - | - | - | - | - | - | - | - | - | - |
| FM7 | - | - | X | - | - | - | - | - | - | X | - | - |
| FM8 | - | - | - | X | - | - | - | - | - | - | - | X |
| T1 | - | - | - | - | - | - | - | - | - | - | - | - |
| T2 | - | - | - | X | X | - | - | - | - | - | X | - |
| T3 | - | - | - | - | X | - | - | - | - | - | X | - |
| T4 | - | - | - | - | - | - | X | X | X | - | - | - |
| T5 | X | - | - | X | - | - | - | - | - | - | - | X |
| T6 | X | - | X | X | - | X | - | X | - | X | - | X |
| T7 | X | X | - | X | - | - | - | - | - | - | - | X |

## Evaluation coverage matrix - continued

| Risk / task | EV28 | EV29 | EV30 | EV31 | EV32 | EV33 |
|---|---|---|---|---|---|---|
| FM1 | - | - | - | - | - | - |
| FM2 | - | - | X | X | X | X |
| FM3 | - | - | - | - | - | - |
| FM4 | - | - | - | - | - | - |
| FM5 | - | - | - | - | - | - |
| FM6 | - | - | - | - | - | - |
| FM7 | X | - | - | - | - | - |
| FM8 | - | X | - | - | - | - |
| T1 | X | X | - | - | - | - |
| T2 | - | - | - | - | - | - |
| T3 | - | - | - | - | - | - |
| T4 | - | - | X | X | X | X |
| T5 | - | X | - | - | - | - |
| T6 | - | X | X | - | X | X |
| T7 | - | X | - | - | - | - |

X means the test is linked to that risk or task. It does not mean the test passed. Complete fixtures and assertions: eval-set.json.

## Evaluation plan

| Test | Layer | Purpose |
|---|---|---|
| EV1 | golden | Valid room-only recommendation |
| EV2 | golden | Wrong tenant cannot read |
| EV3 | golden | Stale operative policy withheld |
| EV4 | golden | Explicit proportional room rule |
| EV5 | golden | Missing room rule is not invented |
| EV6 | golden | Isolated citation check page |
| EV7 | golden | Isolated citation check clause |
| EV8 | golden | Isolated citation check text |
| EV9 | golden | Hidden instructions excluded |
| EV10 | golden | Two concurrent submissions cannot overwrite |
| EV11 | golden | Exact duplicate returns existing record |
| EV12 | golden | Unknown save survives caller expiry and repeated callback |
| EV13 | golden | Cancellation cannot conceal uncertain save |
| EV14 | golden | Missing record proof cannot show saved |
| EV15 | golden | Revoked authority blocks save |
| EV16 | golden | Missing evidence has a visible owned deadline |
| EV17 | golden | Conflicting source evidence abstains |
| EV18 | golden | Semantic duplicate needs staff linking |
| EV19 | offline | Held-out source and arithmetic validation |
| EV20 | online | Queue and save integrity monitoring |
| EV21 | release | Release regression gate |
| EV22 | golden | Policy boundary uses service date, not current date |
| EV23 | golden | Fresh remaining cover caps the recommendation |
| EV24 | golden | Nonzero copay stays with staff |
| EV25 | golden | Changed payload cannot reuse authorization |
| EV26 | golden | Unverified OCR has no authoritative total |
| EV27 | golden | Reviewed rejection remains staff-facing |
| EV28 | golden | Correction requires original record |
| EV29 | golden | Pre-save cancellation has a proven case record |
| EV30 | golden | Half-up rounding is visible at a minor unit |
| EV31 | golden | Negative charge cannot become a draft |
| EV32 | golden | Missing authoritative accumulator version withholds total |
| EV33 | golden | Stale authoritative accumulator version withholds total |

These are test plans. No business controls, provider quality or user outcomes were measured. See eval-set.json for fixtures and pass rules.

## Measurement and rollout

### R1 - Measure examiner time and decision quality

Measure: Median active handling minutes per resolved eligible claim over four weeks; correctness per independently labelled case; zero unauthorized PHI effects

Signals: Quote validity reduces review rework; exact arithmetic protects coverage quality; queue latency protects response time; token cost is operating guardrail

### R2 - Shadow then assist

Phase / owner: Shadow → assist / Claims director

Gate: Shadow entry H1-H4 verified and EV21 synthetic/adapter gate. Exit after two representative weeks: every safety assertion passes, experts inspect all policy/date/room-rule strata and disagreement reasons. Assist entry shadow exit + staffed queue...

### R3 - Test handling-time benefit

Test: ITT on all eligible assigned cases, unresolved included as unresolved; no reallocation of difficult cases. Compare handling minutes and independently reviewed draft quality.

Decision gate: Cluster-randomize examiner teams to assistant/control; stratify by plan, room cap and document type. Keep teams in one arm to reduce contamination. Both...

### R4 - Own rollout and unresolved cases

Phase / owner: Governance / Claims director

Gate: Director signs expansion only after EV21 and H1-H4 evidence; privacy/rules/security owners can veto safety. Operations owns unresolved save tickets across caller expiry, reviews every four hours, escalates to incident lead at deadline...

These are reader summaries. Complete choices, alternatives, gates and owners are in PRD.md and design.json.

## Build handoff and review

Start with build/00-overview.md. Load one milestone file at a time. Exact source rules remain in design.json; do not treat design approval as live write permission.

X1: Implement declared staff-case workflow only. No operational API support or medical/insurance advice is established.

## Review dimensions - evidence in review.json

| Dimension | Score | Dimension | Score |
|---|---|---|---|
| D1 | 3/4 | D7 | 3/4 |
| D2 | 3/4 | D8 | 4/4 |
| D3 | 3/4 | D9 | 3/4 |
| D4 | 3/4 | D10 | 3/4 |
| D5 | 3/4 | D11 | 3/4 |
| D6 | 4/4 | D12 | 3/4 |

IR1 major resolved: Resolved by C2 typed authoritative accumulatorSnapshot, C4 scope/same-request version/timestamp guards, positive fixture source/clock payloads, C7 fresh writer-version query, and isolated missing/stale negatives EV32/EV33. Independently...

IR2 major resolved: Resolved by two complete independently authorized OP-A/OP-B payloads and matching request paths/IDs/hashes/source versions, explicit expectedVersion=3 CAS schedule and one-winner/no-overwrite assertions. Both SHA256 values independently...

## Glossary

canonical: agreed

discriminated: typed

payload binding: approved details

digest: fingerprint

idempotency: duplicate-write protection

reciprocal: two-way

terminal disposition: final outcome

calc_digest: a fingerprint of the approved amount

After delivering DESIGN.pdf, ask the user: Would you also like a visual HTML version in the same ten-stage structure as the app? Generate HTML only after opt-in, unless already requested.

Full assumptions, owners and evidence: assumptions.md. Review method and failed rounds are preserved in review.json and review-round-*.json.